

Graphical User Interfaces (EGUI)

2026L

Consultations: Thursday, 14-16

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Caution: **Only e-mails starting with [EGUI] or [EGUI.A] will be approved!**
all communication regarding the subject should be in private chat to the tutor at TEAMS

Task 3-4: Cinema ticket purchasing system in ASP.MVC+React

- Please write a simple system for purchasing cinema tickets online.
- A new user can register in the system
- The user can specify/change his/her (and the administrator can specify each user):
 - Name and surname
 - Phone number
- The list of cinemas is fixed (i.e. filled in the database using a script) - the cinema has a name and a room size
- Each cinema has one auditorium, the size of which is always a rectangle with n rows and m seats in a row (m and n are different for different cinemas)
- The administrator can create/delete (i.e. not edit) film screenings, deleting a screening deletes all reservations for the selected screening
 - each screening has an assigned cinema
 - each session has a date and time of beginning
 - each screening has a film title
- The user can display the room occupancy for the selected session with free seats marked
- The user can reserve/release seats at a selected screening by indicating the x, y position in the room

Implementation technology

- Tasks 3 and 4 must be completed using ASP.MVC (backend) and React (frontend) technology
- CSS support should be implemented using Bootstrap (or equivalents like Tailwind, Materialize...)

Data format

- The data is stored database. Database vendor and the database structure (design) should be done by student
- Access to the database must be done using Entity Framework

Task 3:

- Implementation of user registration and editing
- Added parallelism when editing a user/deleting a user
- Implementation of creating and deleting sessions

Task 4:

- Implementation of reservation/cancellation of seat reservation at a screening
- Conflict handling (reservation of the same seat by several concurrent users)
- Implementation of displaying occupied seats at a selected screening
- **Preparation of production (1 server) version**

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